# Leon Liu

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## Education

**Carnegie Mellon University** — Pittsburgh, PA
*Bachelor of Science in Computer Science | GPA 3.9/4.0* — *May 2028 (Expected)*

## Experience

**Abundant (YC F24)** — San Francisco, CA
*Software Engineer Intern* — *June 2026 – August 2026*

- Maintained an AI-agent evaluation platform (React, Next.js, FastAPI, PostgreSQL) for benchmarking LLM models and feeding reinforcement learning pipelines at frontier AI labs.
- Optimized experiment run costs and saved company over $2000/week in LLM API calls by migrating from a legacy observability platform and removing deprecated API endpoints.
- Contributed to documentation and novel research of agent failure modes across 10+ large-language models for submission to conferences like NeurIPs and ICLR.
- Designed 500+ task environments to evaluate agents across complex MCP/command line tool usage, container build/configuration, live system-reliability engineering, and post-training, used by companies like Cursor, OpenAI, and Meta.
- Owned 15+ reusable automation skills and CI checks via GitHub Actions to harden model quality-assurance loops and speed iteration on long-horizon agent tasks featured in SWE-Marathon.

**Wayhome** — Pittsburgh, PA (Remote)
*Software Engineer Intern* — *January 2026 – May 2026*

- Designed an AI-driven property listing system for midterm rentals (Ruby on Rails, PostgreSQL), integrating automated listing generation and image analysis to reduce manual user effort.
- Unblocked feature release to 100+ testers via a waitlist and email-distribution pipeline with scheduled Sidekiq-Cron background jobs and fixed window rate limiting.
- Introduced a testing harness adopted by a 12-developer team and enabling rapid testing of cost-heavy features.

**ScottyLabs** — Pittsburgh, PA
*Software Developer, Technical Lead* — *January 2026 – May 2026*

- Led an engineer/designer team maintaining the official Carnegie Mellon Lost & Found platform (Next.js, TypeScript, Prisma, PostgreSQL) serving 16K+ students and processing 100+ items/week.
- Cut page-load times 2x and hosting costs by completing a 3-year-delayed migration (Google Cloud $\rightarrow$ Vercel) and optimizing data loading via pagination and query filtering.
- Hardened the submission and moderation workflow with schemas and guards to reduce invalid submissions $\sim$20%, and shipped a Mailgun pipeline for subscription notifications.

## Projects

**Page-By-Page** - AI Research Summarizer/Recommendation Platform *(React, Tailwind CSS, Flask, PostgreSQL, LangChain)*

- Developed a full-stack research discovery platform that ingests academic papers via the arXiv API and generates LLM-based summaries with personalized recommendations.
- Built backend pipelines for content processing jobs and ranking, improving accessibility of technical research and reducing manual reading time.

**Binder** - Campus Event Logistics App *(Ruby on Rails, SQLite, Stimulus, Docker, GitHub Actions)*

- Maintained an operations app used for Carnegie Mellon's annual Spring Carnival by 70+ coordinators.
- Built interactive admin-style tables with Tabulator and Stimulus, including live client-side countdowns and safe cleanup on navigation to facilitate smooth operation during peak event hours.

**TypoCommand** - Typing Game *(Java, AWT/Swing)*

- Developed a real-time arcade typing game with a custom game loop, double-buffered rendering, event-driven input handling, and low-latency collision detection.
- Improved tester typing speed $\sim$15 WPM on average and was adopted into the curriculum.

## Skills

**Languages:** C++, C, Python, Java, SQL (PostgreSQL, SQLite), TypeScript, JavaScript, Ruby, HTML/CSS
**Frameworks & Web:** React, Next.js, Flask, FastAPI, Ruby on Rails, Express.js, Node.js
**Tools & Cloud:** Git, GitHub Actions (CI/CD), Docker, Vercel, PostgreSQL, MongoDB
**AI/ML**: NumPy, Pandas, scikit-learn, PyTorch, LangChain, Claude Code, Cursor, Codex